

Advanced Method: Bi-Clustering

□ Bi-clustering

- Why and where is Bi-clustering useful?
- Types of Bi-clusters
- Bi-clustering methods
- Enumerating all Bi-clusters using MaPle

Bi-Clustering Methods

- ❑ Bi-clustering: Cluster both objects and attributes simultaneously (treat objects and attributes in a symmetric way)
- ❑ Four requirements:
 - ❑ Only a small set of objects participate in a cluster
 - ❑ A cluster only involves a small number of attributes
 - ❑ An object may participate in multiple clusters, or does not participate in any cluster at all
 - ❑ An attribute may be involved in multiple clusters, or is not involved in any cluster at all

Bi-Clustering Methods

- Example 1: Gene expression or microarray data: a gene sample/condition matrix.
 - Each element in the matrix, a real number, records the expression level of a gene under a specific condition
- Example 2: Clustering customers and products
 - Another bi-clustering problem

	products			
customers	w_{11}	w_{12}	$\cdots$	w_{1m}
	w_{21}	w_{22}	$\cdots$	w_{2m}
	$\cdots$	$\cdots$	$\cdots$	$\cdots$
	w_{n1}	w_{n2}	$\cdots$	w_{nm}

